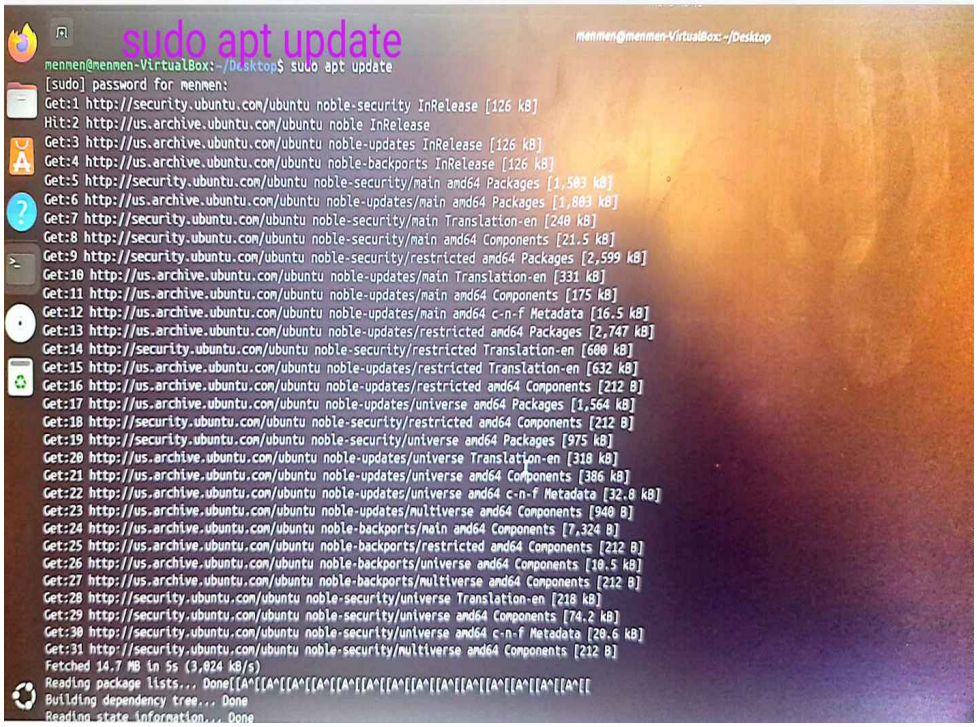
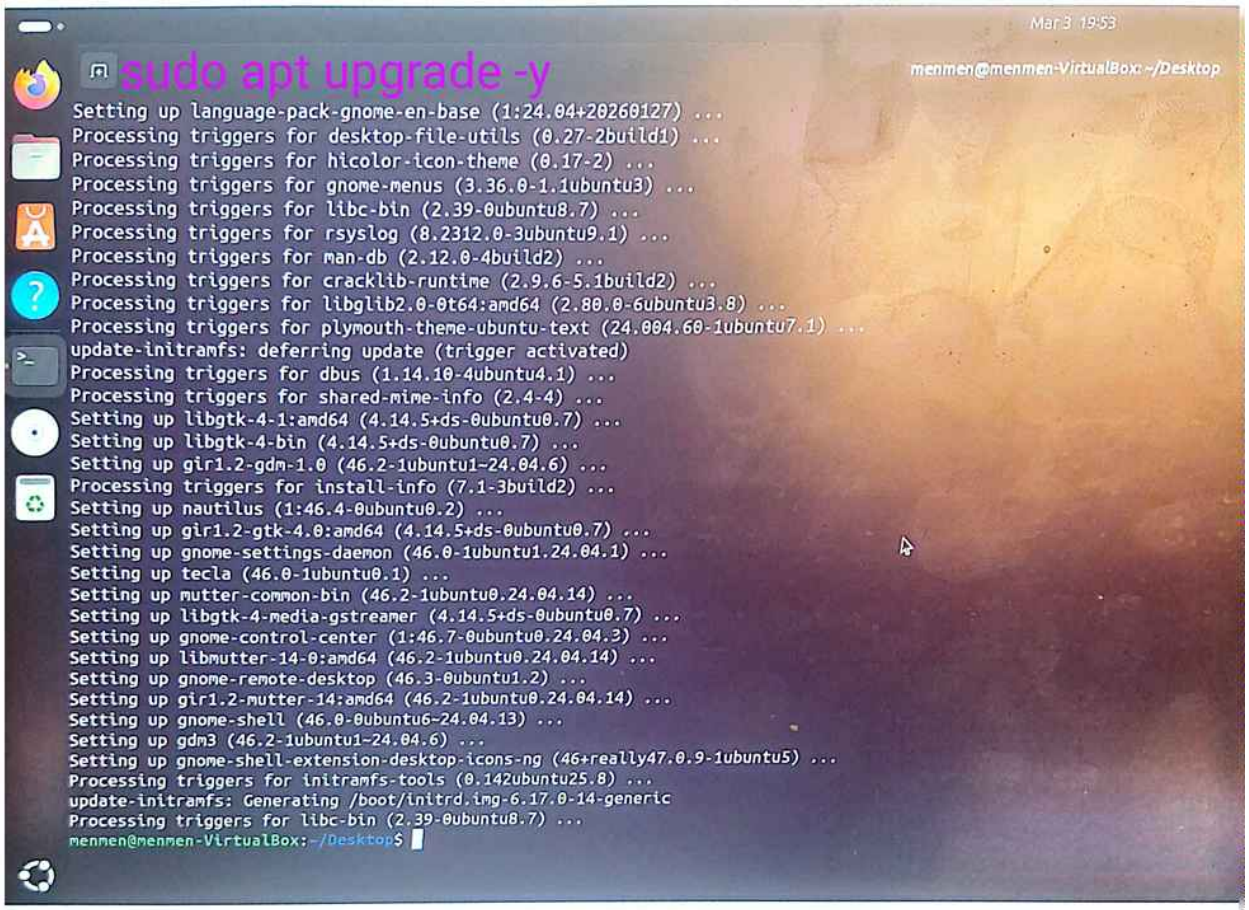
1-General System Tasks:

- **Retrieve available updates: I used: sudo apt update**

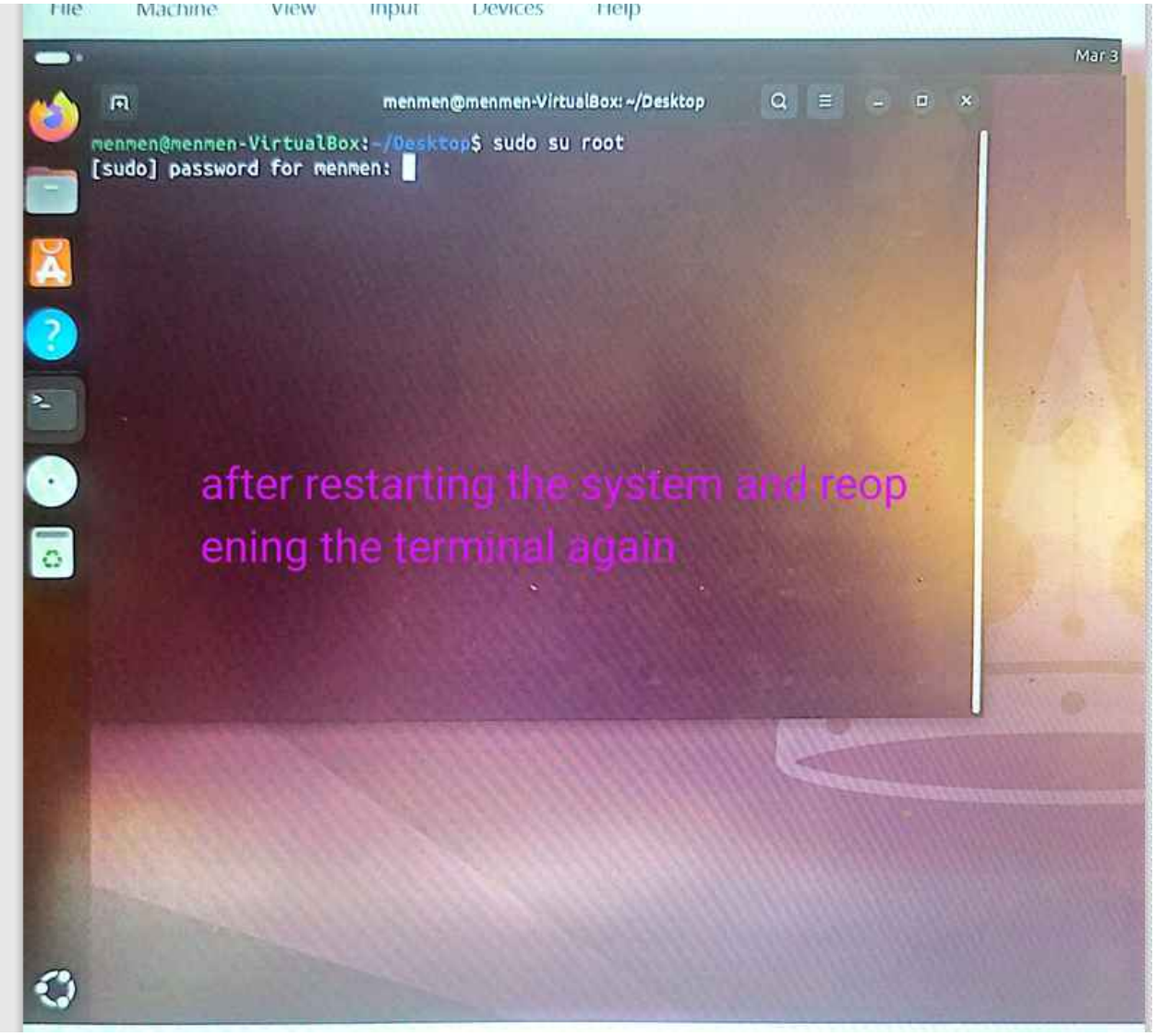


-to upgrade: I used sudo apt upgrade command



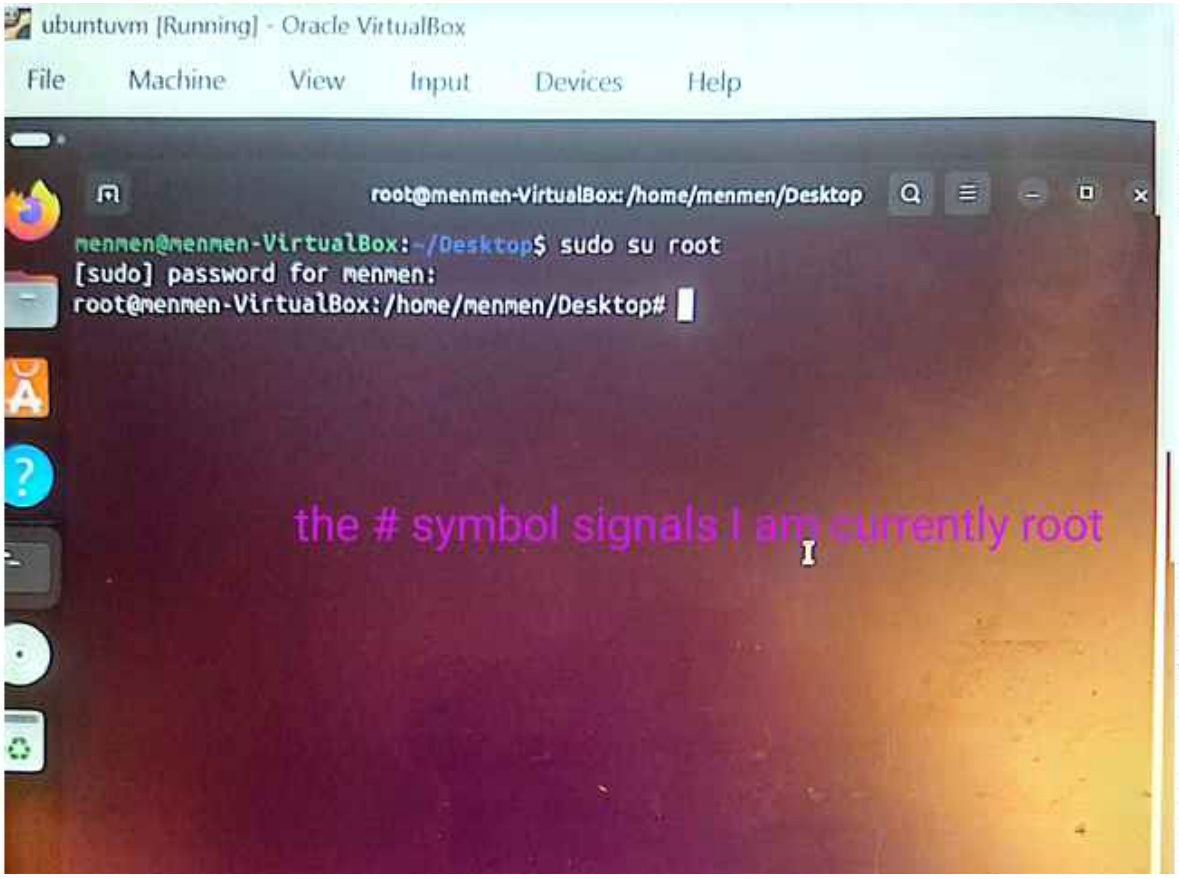
-to reboot: I used sudo reboot command

After I used sudo reboot, it restarted the system, and I reopened the terminal again.



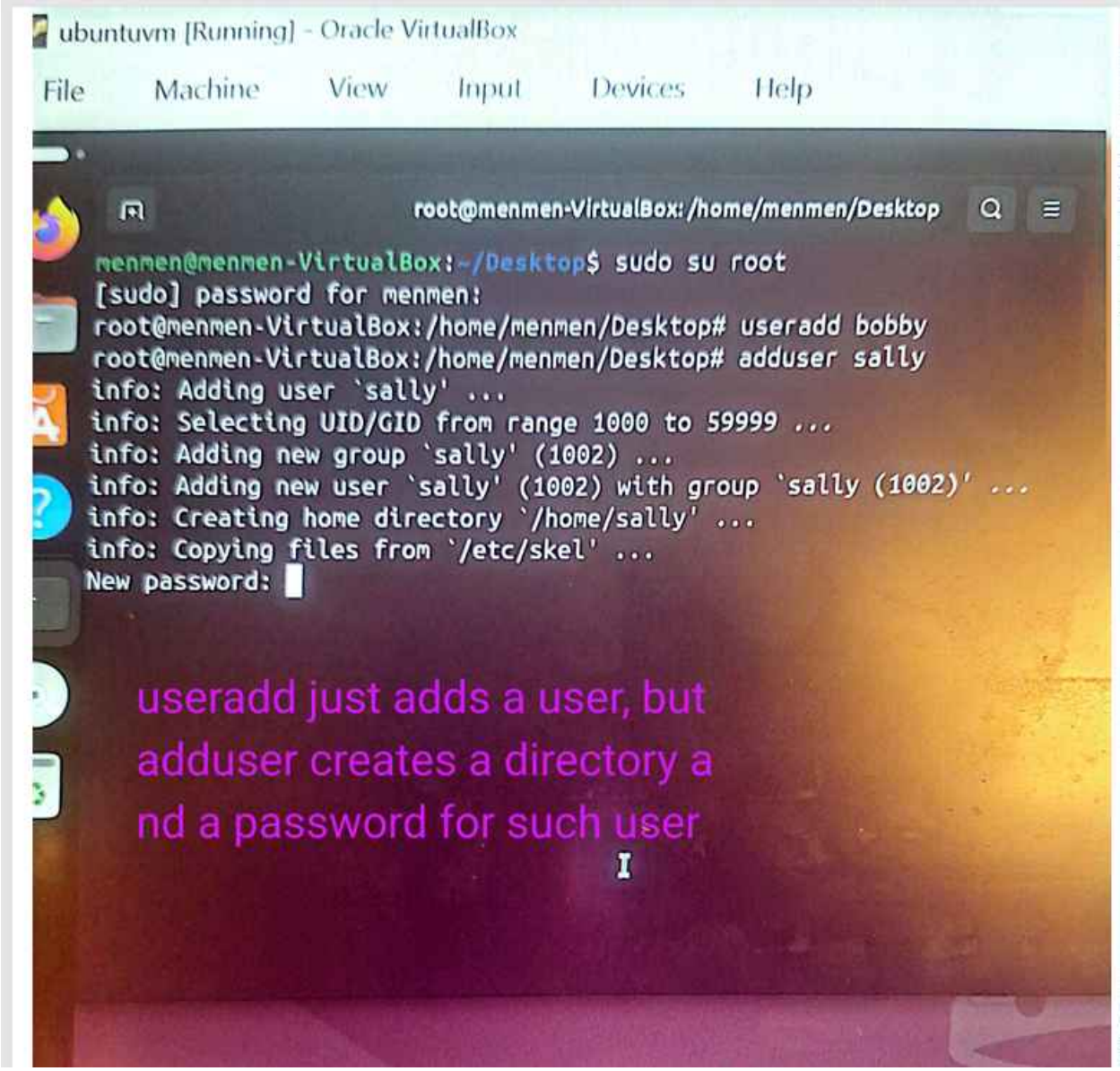
Sudo apt update: Checks the last available versions of ubuntu but does not update anything
Sudo apt upgrade: this one updates the already installed features
Sudo reboot: restarts the system after updating so that the new updates start functioning

User Tasks:
1-Changing the current user to root using: **<u>sudo su root</u>**



-The # symbol signals I am currently root
-Root is like a very powerful user account
-Sudo is short for superuser do which makes you stay yourself as a user but gives you root's power

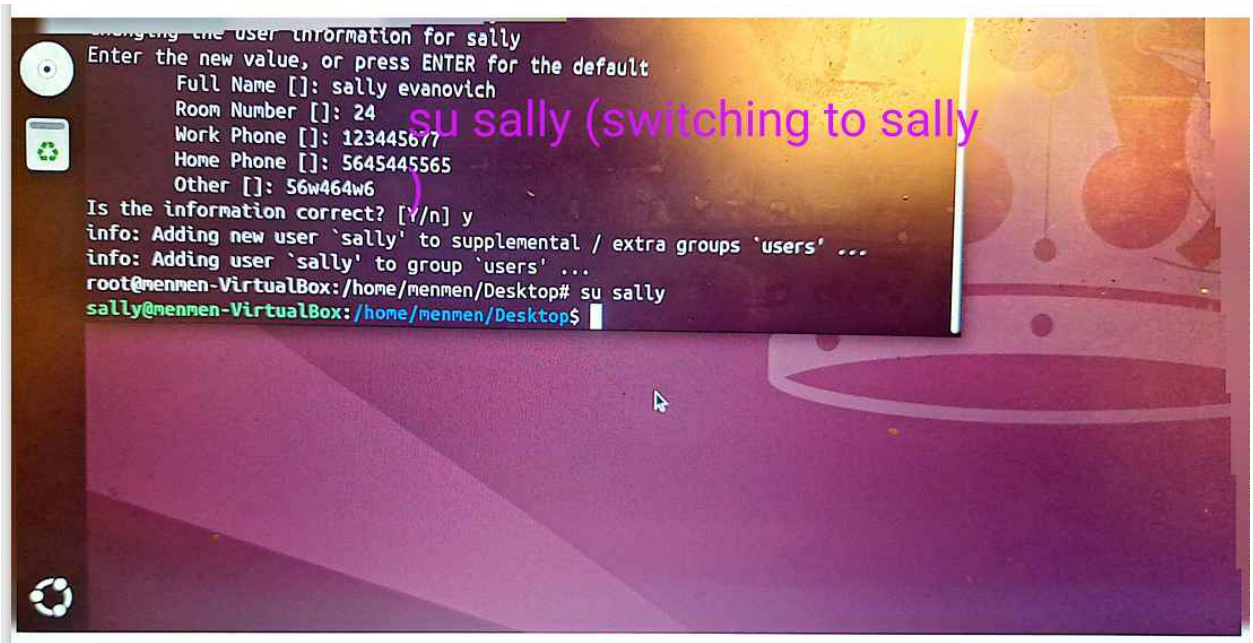
2-creating two new users: Bobby and Sally



Useradd: just creates a user without situp
Adduser: creates a user and asks you to assign password and creates a repo for it

3- Change the current user to sally. What does the prompt look like now?
I ran `su sally` to switch to the sally user. The prompt immediately changed to reflect the new user. It now shows `sally` instead of `menmen` at the beginning. The $ symbol indicates this is a regular user with no root privileges, as opposed

to # which would indicate root access. T



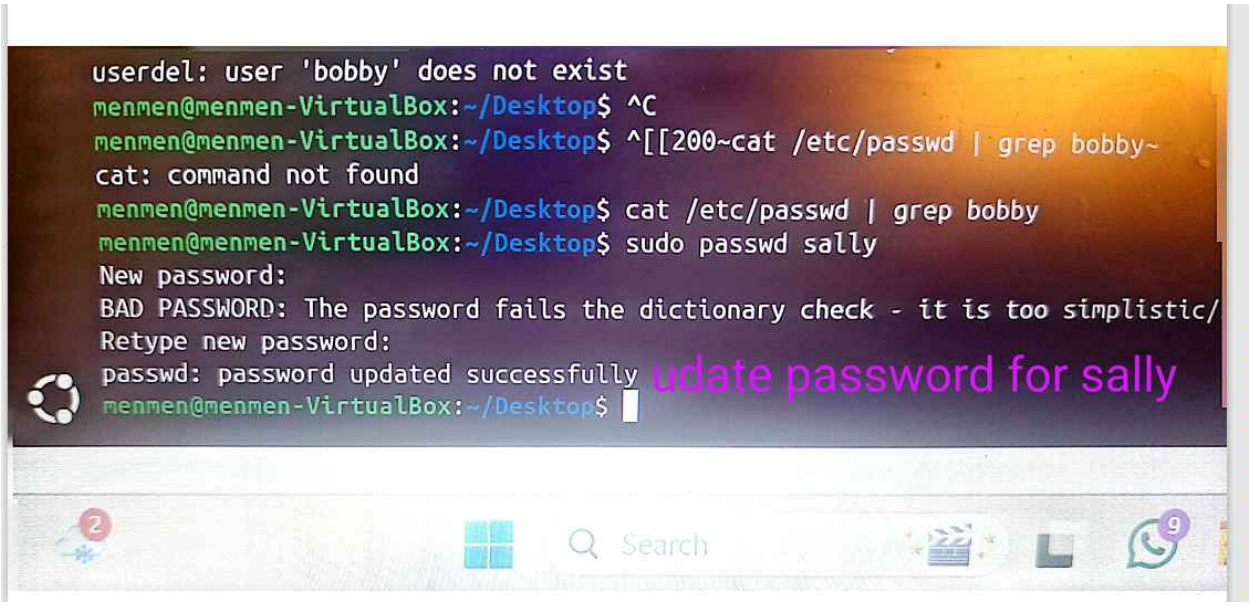
4- While you're logged in as sally still, try to create a new user with the name earl. What happens? Why? What could you do to allow her to create a new user?

I got permission denied after trying to add earl as a user because sally does not have sudo privileges so she cant add new users.

5-Enter exit until you are logged into your own account again. Delete the user bobby.

To delete a user in Linux I Googled 'how to delete a user in Linux' and found the command `userdel`. I ran `sudo userdel bobby` and `sudo userdel -r bobby`, both returned 'user bobby does not exist'. To confirm, I ran `cat /etc/passwd | grep bobby` which returned nothing, verifying bobby was never fully registered in the system. This is because `useradd` without any flags is a low-level command that doesn't always fully create a user the way `adduser` does

6- Change the password of sally to something you can remember using sudo passwd sally



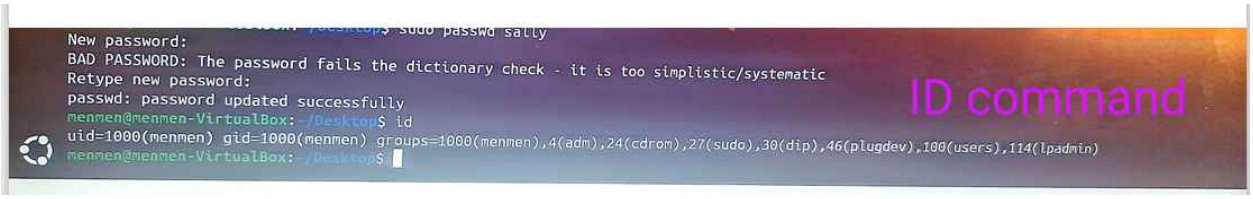
7- Why is it a bad practice to stay logged in as the root?

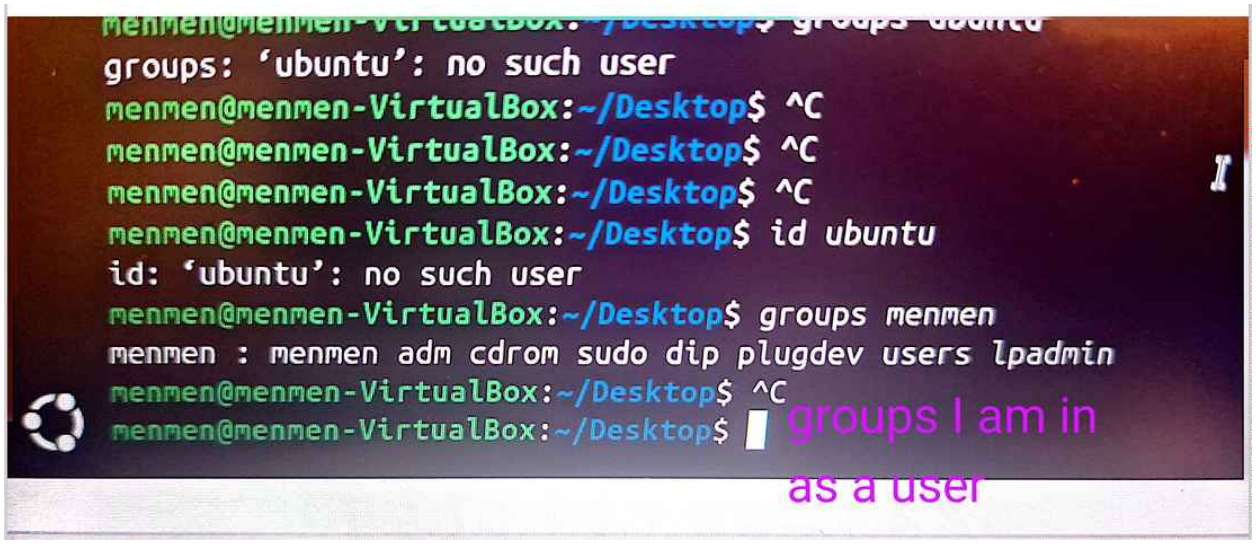
-root has unrestricted access to everything on the system. One wrong command, like typing `rm -rf /`, could destroy the entire system

-If you get hacked or accidentally run malicious software while logged in as root, then it will be worse because the attacker will also have root priviledges

-there is no accountability. When multiple people use a system, staying logged in as root makes it impossible to track which user caused a problem.

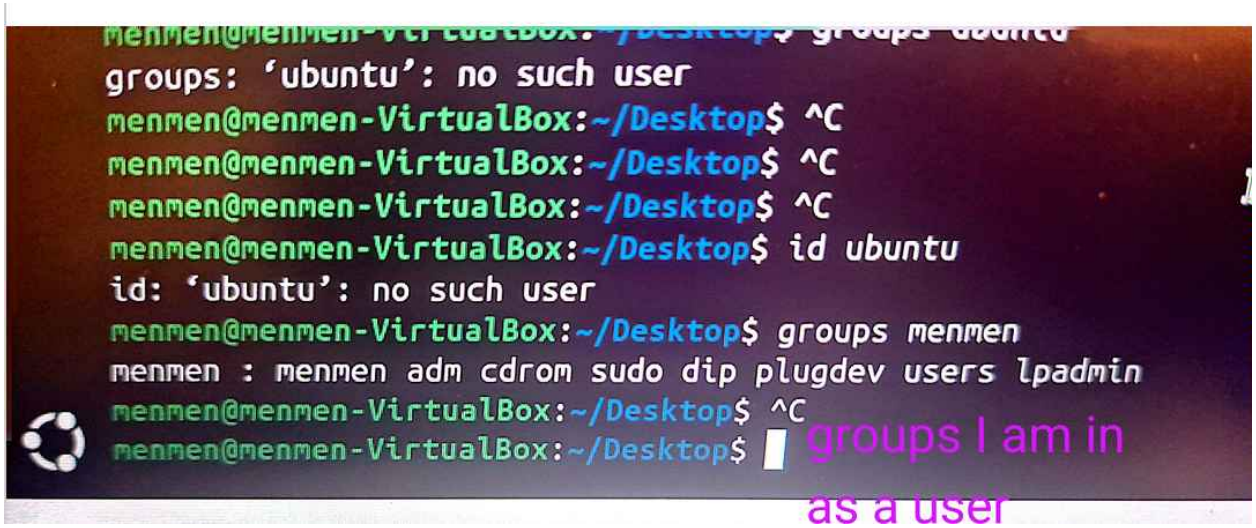
8- Enter the command to see what your user id is.



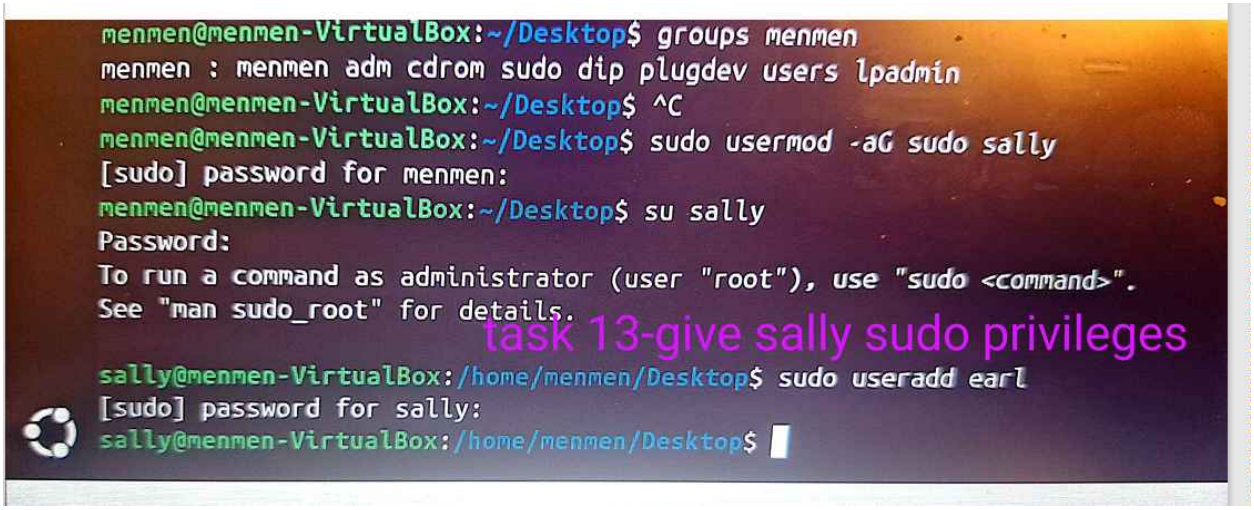


Groups tasks:

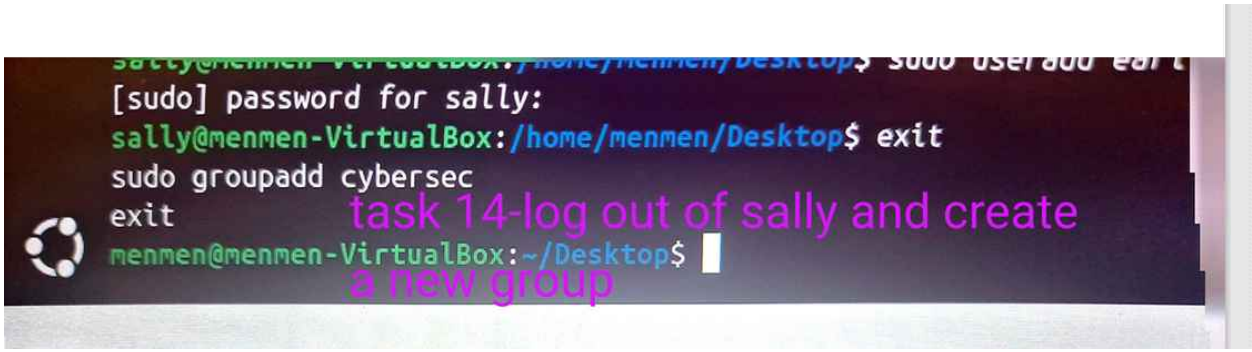
12. What groups does ubuntu belong to?



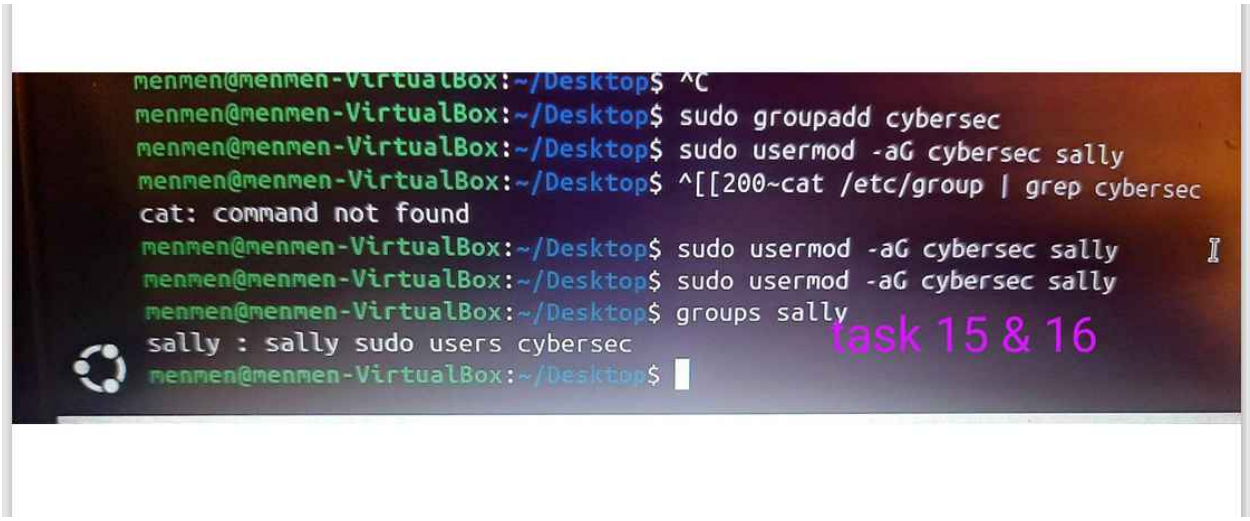
13-13. Give sally the ability to execute sudo commands. Next, try to create a new user while logged in as sally.



14. Log out of Sally and back into your own account. Create a new group called cybersec



15. Add sally to the group, cybersec



16. Check to see which groups sally belongs. What are the various ways to do this?

```
menmen@menmen-VirtualBox:~/Desktop$ ^C
menmen@menmen-VirtualBox:~/Desktop$ sudo groupadd cybersec
menmen@menmen-VirtualBox:~/Desktop$ sudo usermod -aG cybersec sally
menmen@menmen-VirtualBox:~/Desktop$ ^[[200~cat /etc/group | grep cybersec
cat: command not found
menmen@menmen-VirtualBox:~/Desktop$ sudo usermod -aG cybersec sally
menmen@menmen-VirtualBox:~/Desktop$ sudo usermod -aG cybersec sally
menmen@menmen-VirtualBox:~/Desktop$ groups sally
sally : sally sudo users cybersec
menmen@menmen-VirtualBox:~/Desktop$
```

task 15 & 16

```
menmen@menmen-VirtualBox:~/Desktop$ sudo usermod -aG cybersec sally
menmen@menmen-VirtualBox:~/Desktop$ sudo usermod -aG cybersec sally
menmen@menmen-VirtualBox:~/Desktop$ groups sally
sally : sally sudo users cybersec
menmen@menmen-VirtualBox:~/Desktop$ id sally
uid=1002(sally) gid=1002(sally) groups=1002(sally),27(sudo),100(users),1004(cybersec)
menmen@menmen-VirtualBox:~/Desktop$
```

id sally

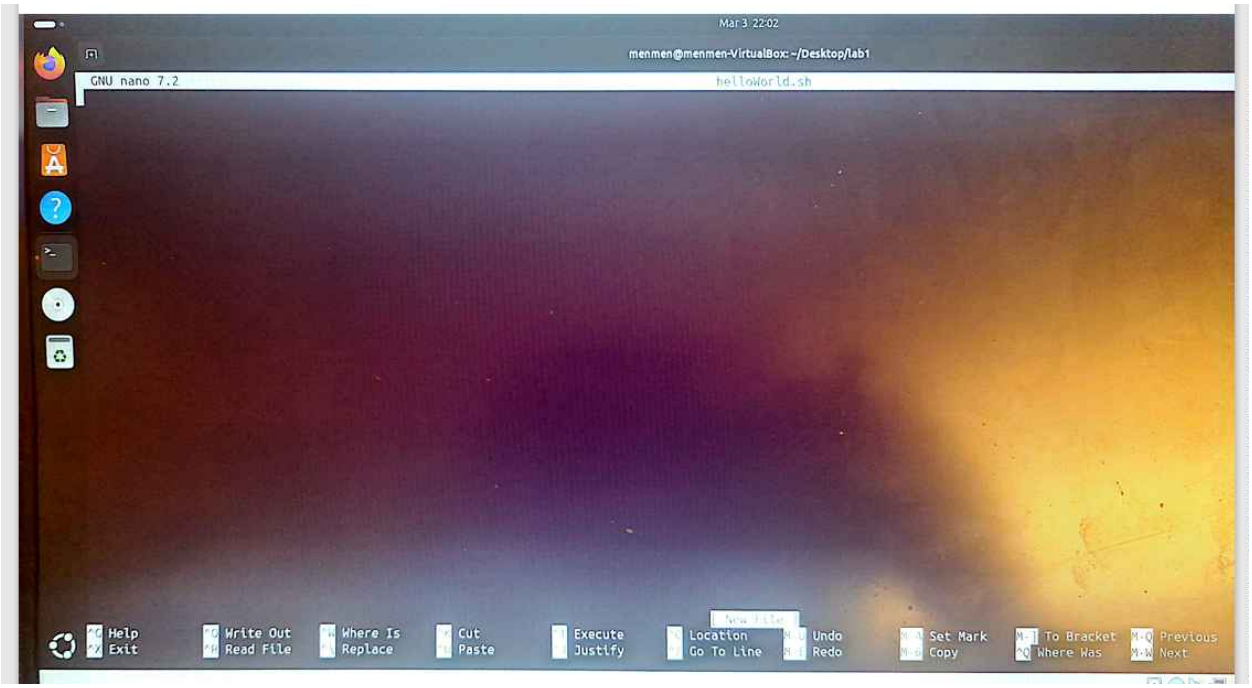
```
menmen@menmen-VirtualBox:~/Desktop$ groups sally
sally : sally sudo users cybersec
menmen@menmen-VirtualBox:~/Desktop$ id sally
uid=1002(sally) gid=1002(sally) groups=1002(sally),27(sudo),100(users),1
menmen@menmen-VirtualBox:~/Desktop$ cat /etc/group | grep sally
sudo:x:27:menmen,sally
users:x:100:menmen,sally
sally:x:1002:
cybersec:x:1004:sally
menmen@menmen-VirtualBox:~/Desktop$
```

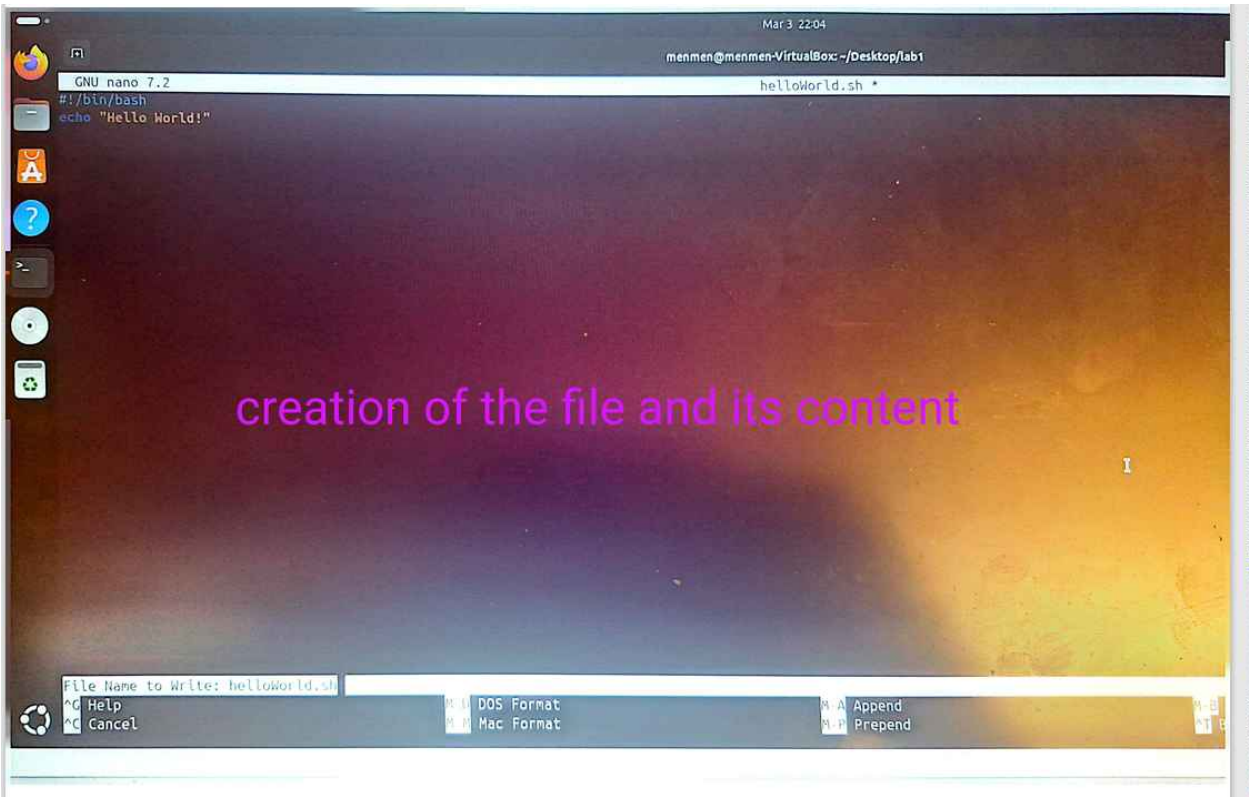
using cat comm
and

17. Create a new directory called lab1. Enter the command to find the permissions of the directory. Who is the owner and group owner of this directory? What permissions does the owner, group and other have

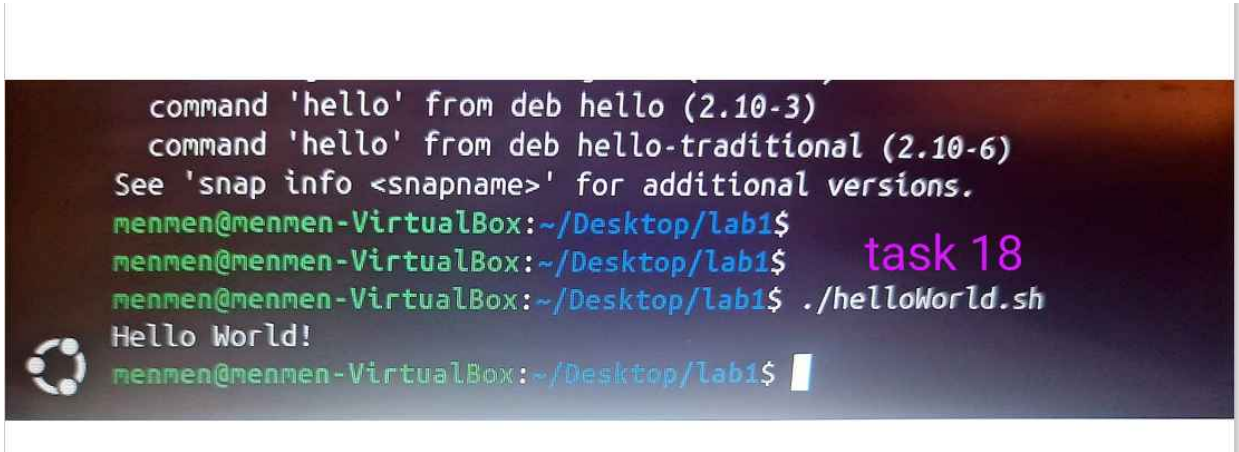
```
sally:x:1002:
cybersec:x:1004:sally
menmen@menmen-VirtualBox:~/Desktop$ mkdir lab1
menmen@menmen-VirtualBox:~/Desktop$ ls -ld lab1
drwxrwxr-x 2 menmen menmen 4096 Mar  3 21:46 lab1
menmen@menmen-VirtualBox:~/Desktop$
```

task 17

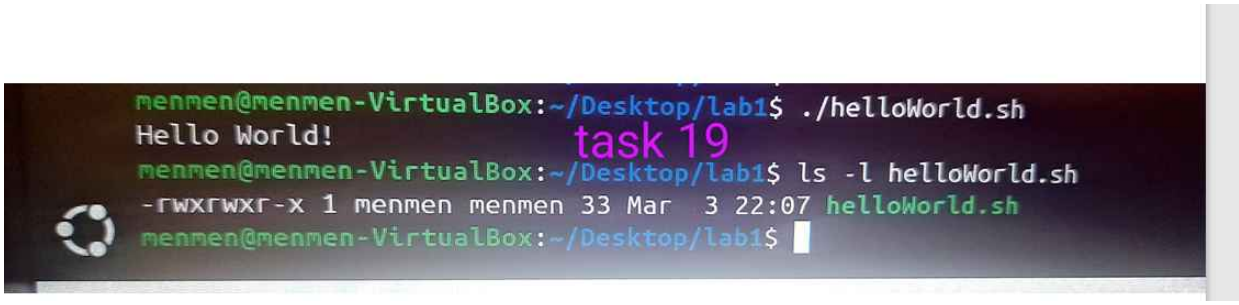




18. Change your directory to lab1. Create a new bash file called, helloWorld. When ran, your program should just print “Hello World!”. (Don’t forget to make your bash f ile executable)



19. Enter the command ls -l helloWorld. What are the reading, writing, and executing permissions for the owner, group and other? a. Change the permissions so the group also has w and x permissions.



20. Use the getfacl command to view the ACL of the file.

21. Using the setfacl command, allow the user, sally, the ability to read and write to the file.

```
menmen@menmen-VirtualBox:~/Desktop/lab1$ sudo setfacl -m u:sally:rw helloWorld.sh
[sudo] password for menmen:
menmen@menmen-VirtualBox:~/Desktop/lab1$ getfacl helloWorld.sh
# file: helloWorld.sh
# owner: menmen
# group: menmen
user::rwx
user:sally:rw-
group::rwx
mask::rwx
other::r-x

menmen@menmen-VirtualBox:~/Desktop/lab1$
```

task 21